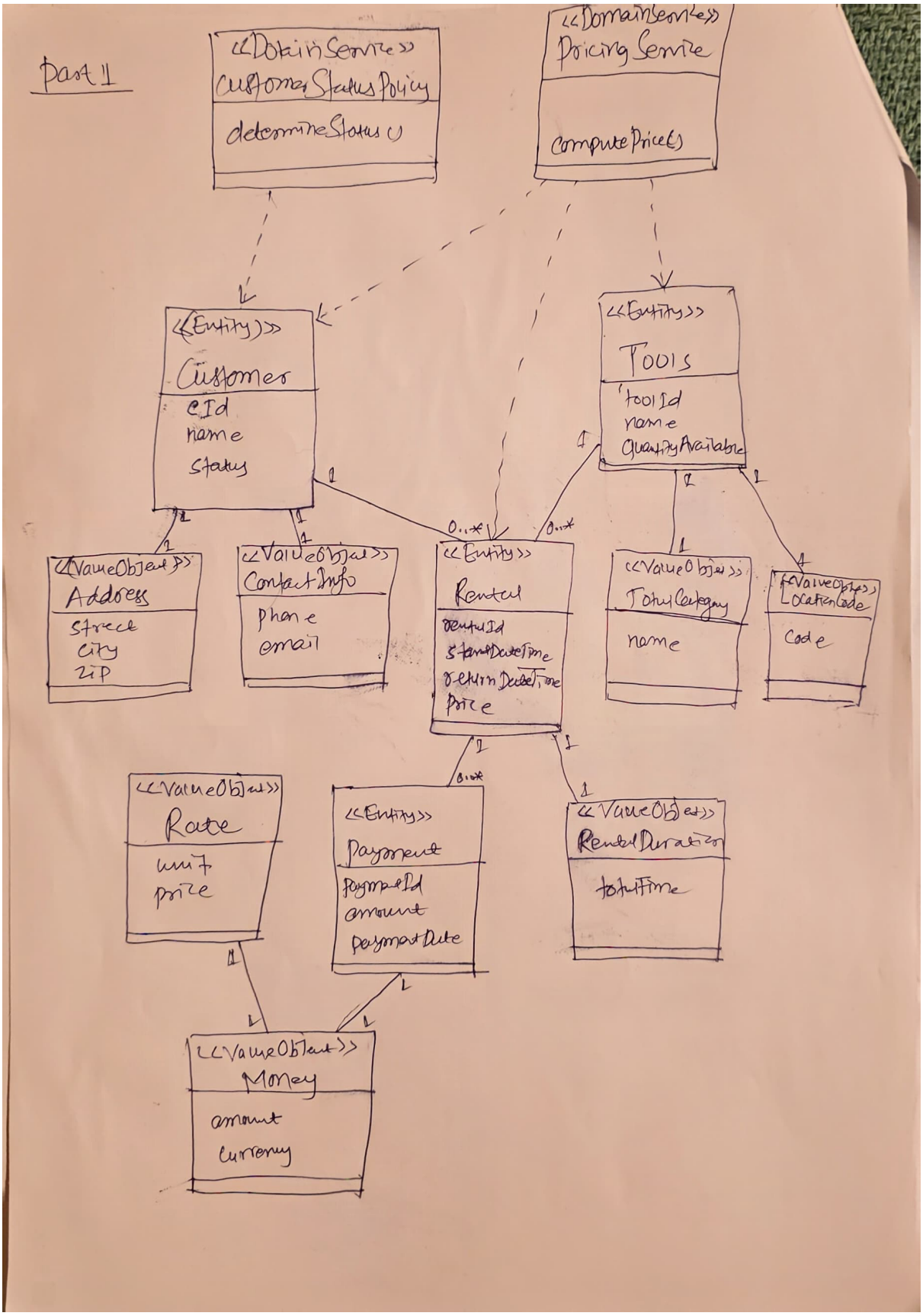
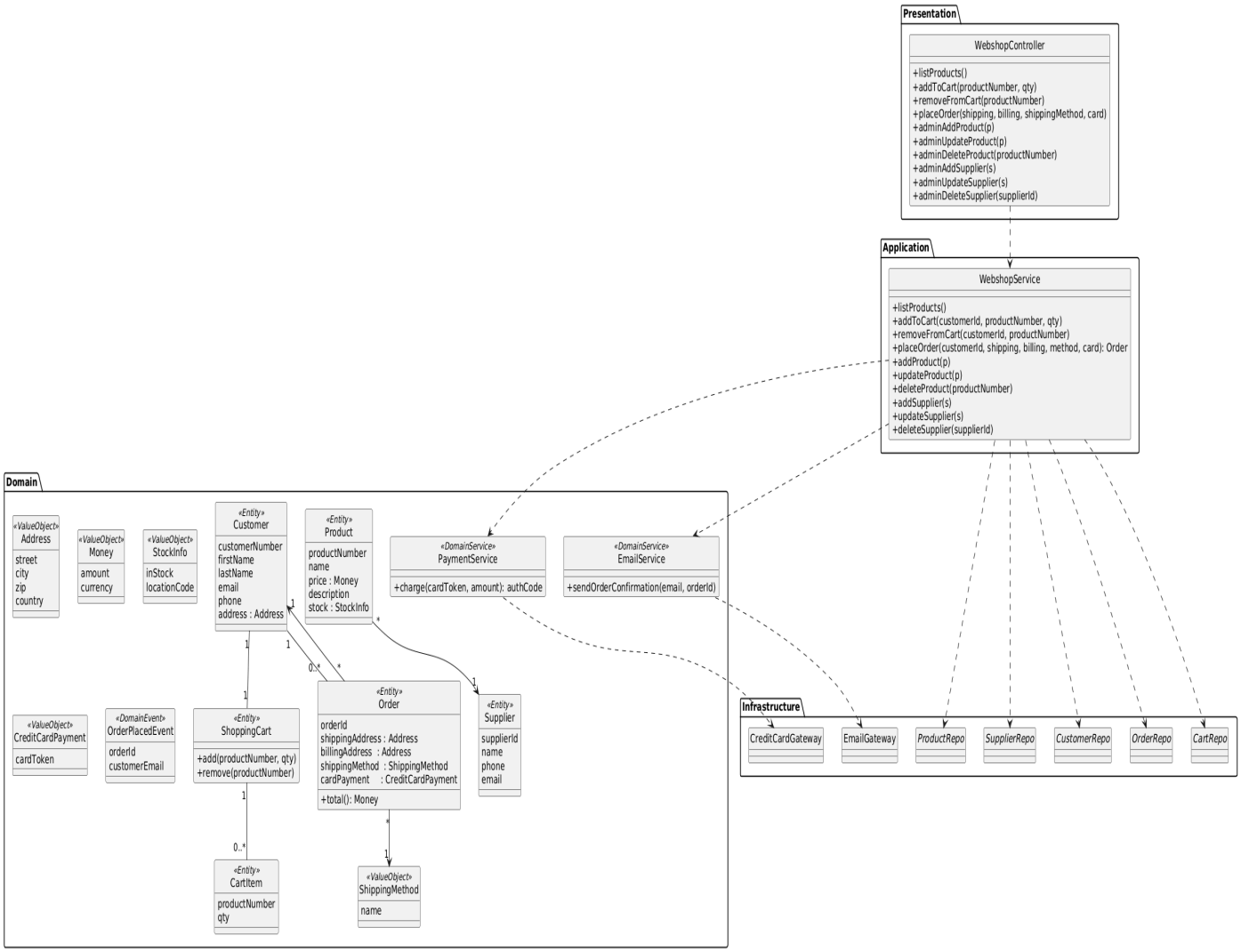
Lab 2 Solution

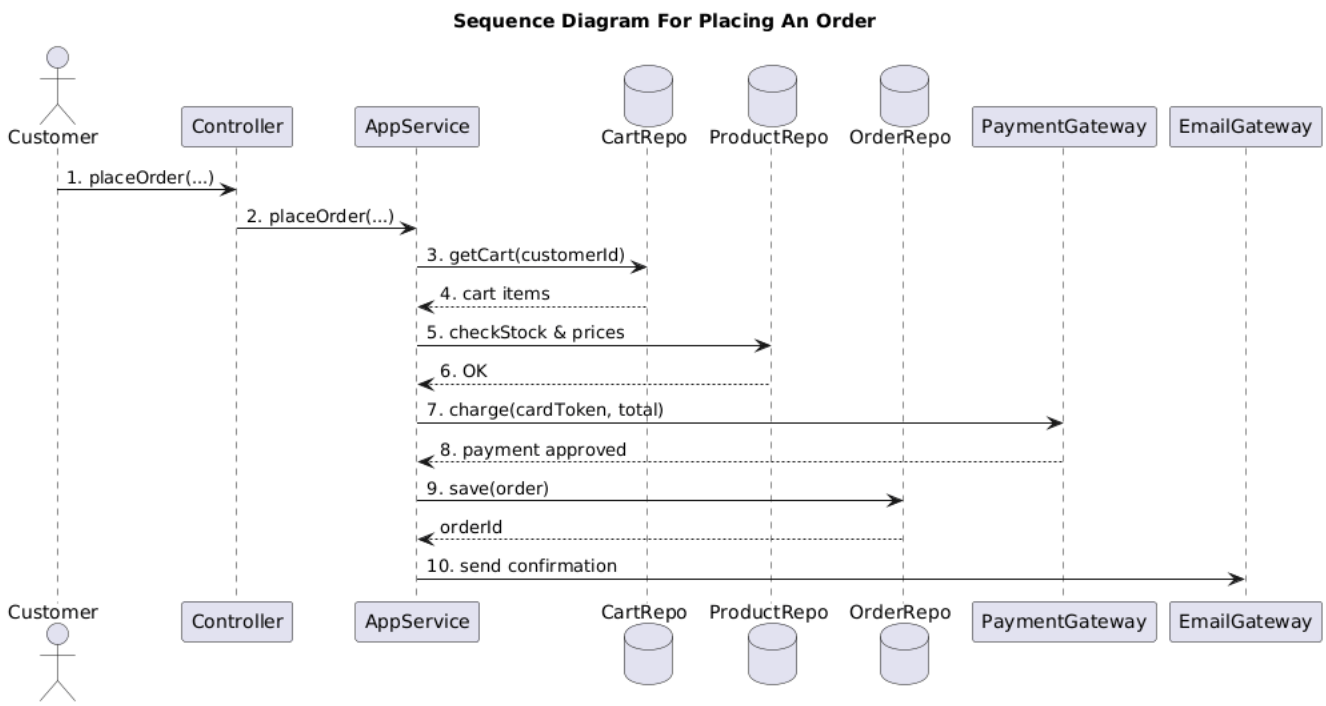
Part 1



Part 2


Presentation
WebshopController
+listProducts()
+addToCart(productNumber, qty)
+removeFromCart(productNumber)
+placeOrder(shipping, billing, shippingMethod, card)
+adminAddProduct(p)
+adminUpdateProduct(p)
+adminDeleteProduct(productNumber)
+adminAddSupplier(s)
+adminUpdateSupplier(s)
+adminDeleteSupplier(supplierId)
Application
WebshopService
+listProducts()
+addToCart(customerId, productNumber, qty)
+removeFromCart(customerId, productNumber)
+placeOrder(customerId, shipping, billing, method, card): Order
+addProduct(p)
+updateProduct(p)
+deleteProduct(productNumber)
+addSupplier(s)
+updateSupplier(s)
+deleteSupplier(supplierId)
Domain
«ValueObject» Address
street
city
zip
country
«ValueObject» Money
amount
currency
«ValueObject» StockInfo
inStock
locationCode
«Entity» Customer
customerNumber
firstName
lastName
email
phone
address : Address
«Entity» Product
productNumber
name
price : Money
description
stock : StockInfo
«DomainService» PaymentService
+charge(cardToken, amount): authCode
«DomainService» EmailService
+sendOrderConfirmation(email, orderId)
«ValueObject» CreditCardPayment
cardToken
«DomainEvent» OrderPlacedEvent
orderId
customerEmail
«Entity» ShoppingCart
+add(productNumber, qty)
+remove(productNumber)
«Entity» Order
orderId
shippingAddress : Address
billingAddress : Address
shippingMethod : ShippingMethod
cardPayment : CreditCardPayment
+total(): Money
«Entity» Supplier
supplierId
name
phone
email
«Entity» CartItem
productNumber
qty
«ValueObject» ShippingMethod
name
Infrastructure
CreditCardGateway
EmailGateway
ProductRepo
SupplierRepo
CustomerRepo
OrderRepo
CartRepo

Part 3


Sequence Diagram For Placing An Order
Customer
Controller
AppService
CartRepo
ProductRepo
OrderRepo
PaymentGateway
EmailGateway
1. placeOrder(...)
2. placeOrder(...)
3. getCart(customerId)
4. cart items
5. checkStock & prices
6. OK
7. charge(cardToken, total)
8. payment approved
9. save(order)
orderId
10. send confirmation